

NATE STEMEN

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SUMMARY

Research engineer experienced in quantum error mitigation, circuit compilation, and translating quantum research into practical tools for current and near-term devices. Lead developer of `mitiq`, a widely-used open-source Python library for error mitigation.

EDUCATION

University of Waterloo MMath in Applied Mathematics 2020–2022

- Thesis: *Quantum Circuit Compilation from the Ground Up* advised by Joel Wallman

New York University B.Sc. in Mathematics and Physics 2013–2017

- Thesis: *An Investigation of Q -Balls* advised by Luciano Medina

EMPLOYMENT

Senior Software Engineer IonQ May 2026–

- Responsible for internal and external error mitigation tooling.

Visiting Researcher QuSoft / University of Amsterdam Jan–March 2026

- Performed computational search for efficient unitary decompositions of bottleneck of QFT on symmetric group.
- Prepped for and made release of first major release of `mitiq` 1.0.0.

Member of Technical Staff Unitary Foundation Mar 2022–Dec 2025

- Technical lead and maintainer of the open-source **Python** library `mitiq` for quantum error mitigation (350k+ downloads, 170+ citations), responsible for core architecture, API design, 20+ releases, and review and integration of contributions from 90+ contributors.
- Designed and implemented a modular, two-step application API for error-mitigation techniques, informed by user interviews, making mitigation circuits and experimental overhead explicit and inspectable.
- Delivered 10+ talks and tutorials at major conferences and research institutions (PyData, SciPy, IEEE QCE, QuSoft), accelerating adoption of `mitiq` by quantum software engineers and researchers.

Software Developer Overleaf 2017–2021

- Improved $\text{\LaTeX}$ autocomplete using statistical analysis of 40,000 open-source documents, enhancing user experience for 300,000+ daily users.
- Monitored and supported data migration from PostgreSQL to MongoDB.

Summer Researcher New York University 2016

- Used **Python** to numerically solve nonlinear Schrödinger equations modeling electromagnetic pulse propagation in nonlinear media.

Summer Researcher Yale University (PROSPECT Experiment) 2014 & 2015

- Built an optical simulation in **C++** to optimize detector design and study light collection and uniformity.
- Implemented pulse-shape discrimination techniques in **Python** to improve neutrino event selection.

PUBLICATIONS

1. LaRose, R. et al. (Aug. 2022). Mitiq: A software package for error mitigation on noisy quantum computers. *Quantum* 6, p. 774. URL: <https://doi.org/10.22331/q-2022-08-11-774>.
2. McDonough, B. et al. (2022). “Automated quantum error mitigation based on probabilistic error reduction”. In: *2022 IEEE/ACM Third International Workshop on Quantum Computing Software (QCS)*, pp. 83–93. arXiv: [2210.08611](https://arxiv.org/abs/2210.08611) [quant-ph].
3. Ashenfelter, J. et al. (2016). Background Radiation Measurements at High Power Research Reactors. *Nucl. Instrum. Meth. A* 806, pp. 401–419. arXiv: [1506.03547](https://arxiv.org/abs/1506.03547) [physics.ins-det].
4. Ashenfelter, J. et al. (2015). Light Collection and Pulse-Shape Discrimination in Elongated Scintillator Cells for the PROSPECT Reactor Antineutrino Experiment. *JINST* 10.11, P11004. arXiv: [1508.06575](https://arxiv.org/abs/1508.06575) [physics.ins-det].

SERVICE

Graduate Student Mentor	University of Amsterdam	2025 & 2026
Graduate Student Mentor	University of Washington	2024 & 2025
IEEE QCE 2025 Workshop organizer	Quantum Software 2.1	2025
WERQSHOP Chair Organizer	https://werq.shop	2025
Quantum Computing Devroom Chair	FOSDEM	2025
SciPy 2025 Reviewer		2025
QED-C mentor		2023–2024
Equity, Diversity and Inclusion Committee	University of Waterloo; IQC	2021–2022
Strategic Plan Implementation Working Group	University of Waterloo	2021

CONTINUING

EDUCATION

Advanced Representation Theory and Applications	University of Amsterdam (audit)	Feb–Mar 2026
Advanced Quantum Algorithms	University of Amsterdam (audit)	Feb–Mar 2026
Hands-on quantum error correction with Google Quantum AI	Coursera	Feb 2026
CSE 599C: Quantum Learning Theory	University of Washington (audit)	Jan–Mar 2025
CSE 534: Quantum info. and computation	University of Washington (audit)	Sep–Dec 2024
Quantum Machine Learning Workshop	QSciTech-QuantumBC	Jan–Feb 2022
Presenting Data and Information	Edward Tufte	Nov 2019

TOOLS

Languages

- Python, JavaScript, SQL, Ruby, bash

Software

- git/GitHub, docker, Linux, MacOS, L^AT_EX

Quantum

- SDKs: Cirq, Qiskit, pyQuil, Qibo